Junsheng Huang

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Education

University of Illinois at Urbana-Champaign Aug. 2025 – May 2027 (expected)

Master of Science in Electrical and Computer Engineering (GPA: 3.91 / 4.00)

- **Relevant Courses:** Reinforcement Learning (ECE586), Natural Language Processing (CS546), Computer Vision (CS543), Graph Machine Learning (CS598)

Zhejiang University-University of Illinois at Urbana-Champaign Aug. 2021 - Jun. 2025

Dual Bachelor's Degrees in Electrical and Computer Engineering (GPA: 3.96 / 4.00)

- **Relevant Courses:** Computer Systems Engineering (ECE391), Data Structures (CS225), Artificial Intelligence (CS440), Algorithms (CS374), Machine Learning (CS446), Applied Parallel Programming (ECE408)

Research Interests

My research focuses on how LLMs and autonomous agents can learn and improve from feedback and experience. I am particularly interested in post-training methods for improving reasoning and decision-making, as well as recursive self-improvement (RSI) that enables agents to learn from multi-step interactions and experimental outcomes. My long-term goal is to develop autonomous agents that can advance scientific discovery and solve complex engineering problems.

Publications

- **Junsheng Huang***, Yifan Sun*, Zhouer Zhang*, Ning Wei, Yening Liu, Michael Molter, Pavan Kumar Hanumolu, Elyse Rosenbaum, Bin Hu, Huan Zhang. *SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design*. Accepted to **NeurIPS** 2026 Main Conference.
- **Junsheng Huang**, Zhitao He, Yuchen Huang, Sandeep Polisetty, Qingyun Wang, Yi R. Fung. *MAC-Tuning: LLM Multi-Compositional Problem Reasoning with Enhanced Knowledge Boundary Awareness*. Accepted to **EMNLP** 2025 Main Conference.
- Jason Vega, **Junsheng Huang**, Gaokai Zhang, Hangoo Kang, Minjia Zhang, Gagandeep Singh. *Stochastic Monkeys at Play: Random Augmentations Cheaply Break LLM Safety Alignment*. Preprint.
- W. Yan, Z. Deng, M. Molter, R. Krishna, N. Rouf, S. S. Sofla, E. Thieme, S. Chatterjee, P. P. P. Linn, Y. Wen, X. Xiao, Z. Zhang, **Junsheng Huang**, Y. Liu, Y. Zhou, P. Franzon, M. Swaminathan, E. Rosenbaum, and A. Aysu. "From Predict to Generate: A Cross-Layer Survey of Generative AI for IC Design and Manufacturing." Preprint.

Research Experience

Evaluation and Self-Improvement of Autonomous Research Agents (Ongoing Research) | Advisor: *Chao Zhang*

- Investigating how autonomous research agents can improve their research capabilities through iterative experimentation and feedback.
- Developing evaluation protocols to assess research outcomes, intermediate decision quality, and the generalizability of agent self-improvement.
- Exploring hypothesis evaluation and feedback-driven experiment selection to improve the reliability and efficiency of autonomous research.

SAFTAC: Simulation-Augmented Fine-Tuning of Open-Source LLMs for Analog Circuit Design | Advisor: *Huan Zhang*

- Led the development of SAFTAC, a simulation-augmented post-training framework for specification-conditioned analog circuit generation and feedback-guided correction.
- Built an automated data curation pipeline integrating LLM annotation, Bayesian optimization, and SPICE simulation, constructing 8,626 design tasks across 10 circuit types.
- Designed a three-stage SFT curriculum followed by GRPO-based reinforcement fine-tuning, leveraging simulation-grounded rewards to train both circuit generation and feedback-guided correction.
- Fine-tuned Qwen3-8B, achieving 71.4% single-pass and 78.2% two-pass pass@1, outperforming the strongest closed-source baselines by 15.9 and 7.8 percentage points, respectively.

MAC-Tuning: LLM Multi-Compositional Problem Reasoning with Enhanced Knowledge Boundary Awareness
Advisor: *Yi R. (May) Fung*

- Led research on LLM confidence estimation and knowledge-boundary awareness in multi-problem settings, covering both independent and sequentially related questions.
- Proposed MAC-Tuning, a two-stage supervised fine-tuning framework that constructs multi-question QA and confidence data to decouple answer prediction from confidence estimation.
- Evaluated MAC-Tuning across six benchmarks and multiple LLM families and sizes, improving average precision by up to 25% over baselines while reducing confidence calibration error.

Random Augmentations Cheaply Break LLM Safety Alignment | Advisor: *Gagandeep Singh*

- Investigated the robustness of LLM safety alignment to low-cost random prompt augmentations across 17 models, examining the effects of model size, quantization, fine-tuning-based defenses, and decoding strategies.
- Implemented character-level perturbations and random string insertion strategies with varying augmentation strengths to evaluate their effectiveness in bypassing safety alignment.
- Co-designed human-calibrated safety evaluation metrics using 1,220 manually annotated cases, demonstrating that just 25 random augmentations can substantially increase attack success rates on safety-aligned LLMs.

Technical Skills

Programming Languages: Python, C/C++, CUDA, x86 Assembly, MATLAB
Frameworks & Tools: PyTorch, Hugging Face Transformers, PEFT/LoRA, DeepSpeed/FSDP, vLLM, verl
Spoken Languages: English (Proficient), Mandarin (Native), Cantonese (Native)

Honors

Honorable Mention, Mathematical Contest of Modeling (MCM) (**May 2023**)

ZJU-UIUC Institute Dean's List, **Sophomore Year**

ZJU-UIUC Institute Third-Class Academic Excellence Award, **Sophomore and Junior Years**

UIUC Grainger Engineering Department Dean's List, **Junior Year**

Teaching and Leadership

Teaching Assistant for ECE484 (Principles of Safe Autonomy) Aug 2026 - Dec 2026
Teaching Assistant for CS446/ECE449 (Machine Learning) Jan 2026 - May 2026
Teaching Assistant for MATH285 (Introduction to Differential Equations) Jan 2025 - Jun 2025
Teaching Assistant for MATH241 (Calculus III) Sep 2024 - Dec 2024
Course Assistant for ECE391 (Computer System Engineering) Jan 2024 - May 2024
Student Representative for ECE Major in ZJUI Sep 2021 - Jun 2022, Sep 2024 - Jun 2025